Subject Description Form

Subject Code	AAE6203
Subject Title	Mathematics for Aircraft Structure, Guidance, Navigation, and Control
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with understanding and knowledge about the key mathematics in aircraft structure, guidance, and control. 2. To develop students' capability to numerically analyze research problems from a mathematical view, for example using the matrix to represent the problem. 3. To provide students with in-depth mathematical examples of aircraft structure, guidance, and control.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. obtain in-depth knowledge of basic matrix concepts, notations, vectors, matrix space, and key properties of the matrix, solving the optimization problem using the matrices. b. competently apply fundamental mathematic concepts to aircraft structure, guidance, and control problems. c. critically evaluate the characteristics of the given engineering problem using the properties of the matrix. d. identify the key challenges of the research in aircraft structure, guidance, and control from the mathematical view.
Subject Synopsis/ Indicative Syllabus	1. **Basic topology**: Finite, countable, and uncountable sets; metric spaces; compact sets; perfect sets; connected sets. 2. **Matrix analysis**: Eigenvalues, eigenvectors, and similarity; definition of norms and inner products; properties of norms; matrix norms; singular value decomposition; Schur decomposition; the Jordan canonical form theorem. 3. **Optimization**: Convexity; affine and convex sets; convex function; LP weak and strong duality; optimality conditions. 4. **Dynamic System and Stability**: Application to linear systems theory and stability. 5. **Case Studies**: Optimal control, linear quadratic regulator, an air traffic control problem.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures/tutorials and homework assignments. 2. The lectures/tutorials aim at providing students with integrated knowledge of mathematics in aircraft structure, guidance, and control. In-class case studies will be raised to develop student's skills in applying mathematical concepts to real engineering problems. 3. Homework assignments are used to allow students to reflect on and deepen their knowledge of a selected topic.

Teaching/Learning Methodology	Intended subject learning outcomes			
	a	b	c	d
1. Lectures/tutorials	√	√	√	√
2. Homework assignments	√	√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Assignments	30%	√	√	√	√
2. Midterm exam	20%	√	√	√	√
3. Final exam	50%	√	√	√	√
Total	100%				

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

1. The assessment is comprised of 50% continuous assessment and 50% final exam.

2. The continuous assessment consists of assignments and midterm exam. They are used to evaluate the progress of students' study, assist them in self-monitoring of fulfilling the respective subject learning outcomes, and enhance the integration of the knowledge learned.

3. The final exam is used to assess the knowledge acquired by the students for understanding and analyzing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.

Student Study Effort Expected		
	Class contact:	
	▪ Lectures	33 Hrs.
	▪ Tutorials	6 Hrs.
	Other student study effort:	

	▪ Self-study	33 Hrs.
	▪ Projects/homework assignments	50 Hrs.
	Total student study effort	122 Hrs.
Reading List and References	1. W. Rudin, *Principles of mathematical analysis*, 3d ed. New York: McGraw-Hill, 1976. 2. R. A. Horn and C. R. Johnson, *Matrix analysis*, 2nd ed. Cambridge; Cambridge University Press, 2013. 3. S. P. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge; Cambridge University Press, 2004.	

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